Innovating Strategies for Enhancement of Production Capacity of CLW at PBT

Presented By
Kazi Shafi Sami
Assistant Mechanical Engineer (Probationer)
Bangladesh Railway

Content

- Introduction
- Objective and Major Functions of CLW
- CLW at a Glance
- Working Procedure of GOH
- Critical Path Method
- Sections at CLW
- Challenges of CLW
- Strategy, Action and Expected Outcome
- Implementation Roadmap
- Conclusion

Introduction

- The Central Locomotive Workshop, commonly referred to as CLW, is the principal facility for locomotive maintenance in Bangladesh.
- The workshop specializes inGeneral overhauling of both Meter Gauge and Broad Gauge Diesel Electric Locomotives, ensuring their long-term reliability and efficiency.
- By implementing strategic maintenance practices and modernization initiatives, CLW plays a vital role in strengthening the performance and sustainability of Bangladesh Railway’s locomotive fleet.

Objective and Major Functions of CLW

Objective of CLW:

To achieve reliable performance of Locomotive, each Locomotive goes under General Overhauling(GOH) after six years in the Central Locomotive Workshop (CLW)

Major Functions:

- G.O.H. (General Overhauling): BG and MG locomotives undergo general overhauling after every 06 years.
- Special Repair of damaged Locomotives in accidents and complex repair works which are not possible in other diesel locomotive workshops

CLW at a Glance

CLW at a Glance

Workshop in Operation	May 14, 1992
Workshop Complex	111 Acre (Workhop Area-28 Acre)
Approved Manpower	728
Available Manpower	184 (26 percent of Approved Manpower)
Division	8 (Mechanical, Stores, Security, Electrical, Accounts, Engineering, Medical & Training Unit)
Total Number of Plants & Machinery	362
KPI(Key Point Installation) Class	1C

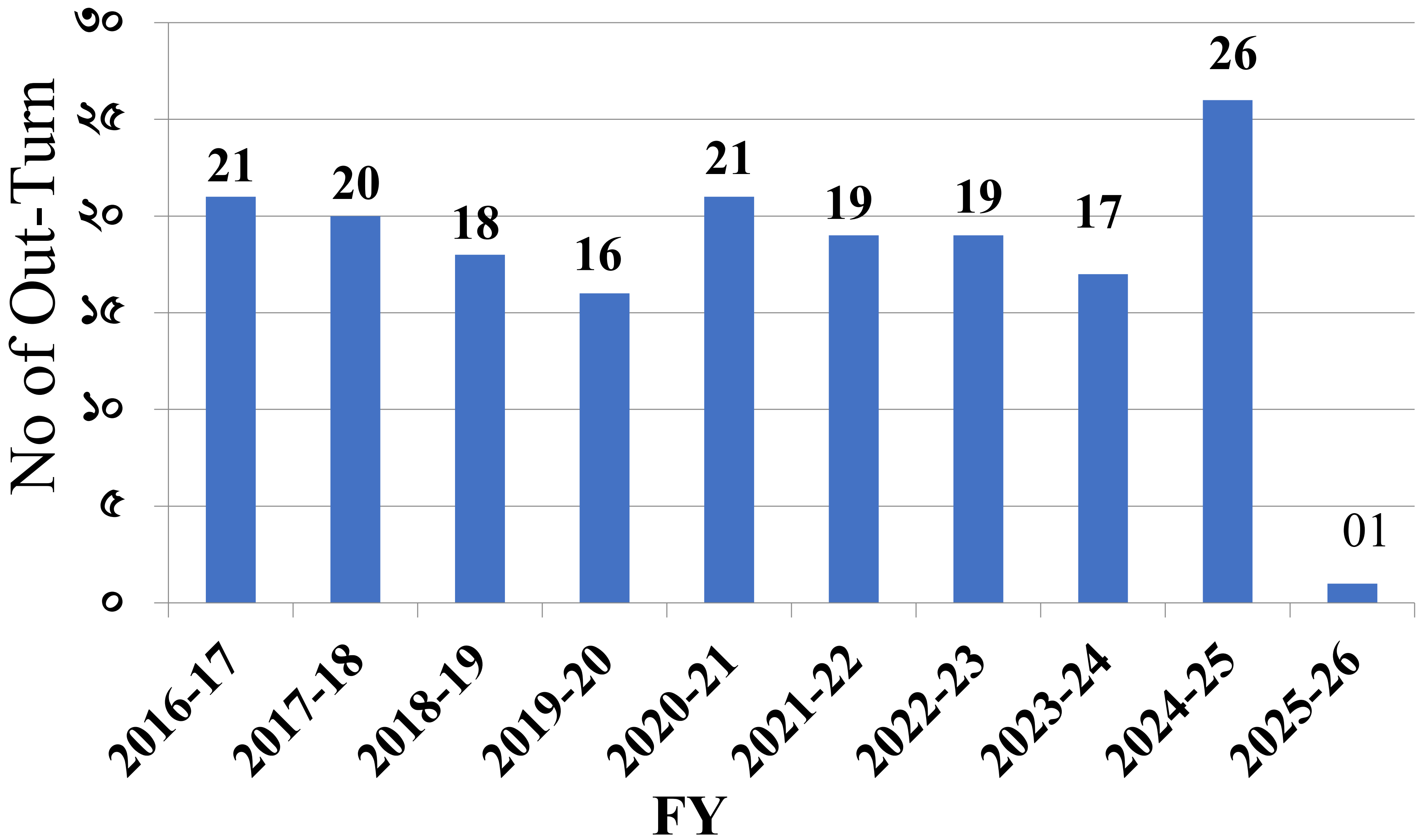
Location

District	Upazilla
Dinajpur	Parbatipur

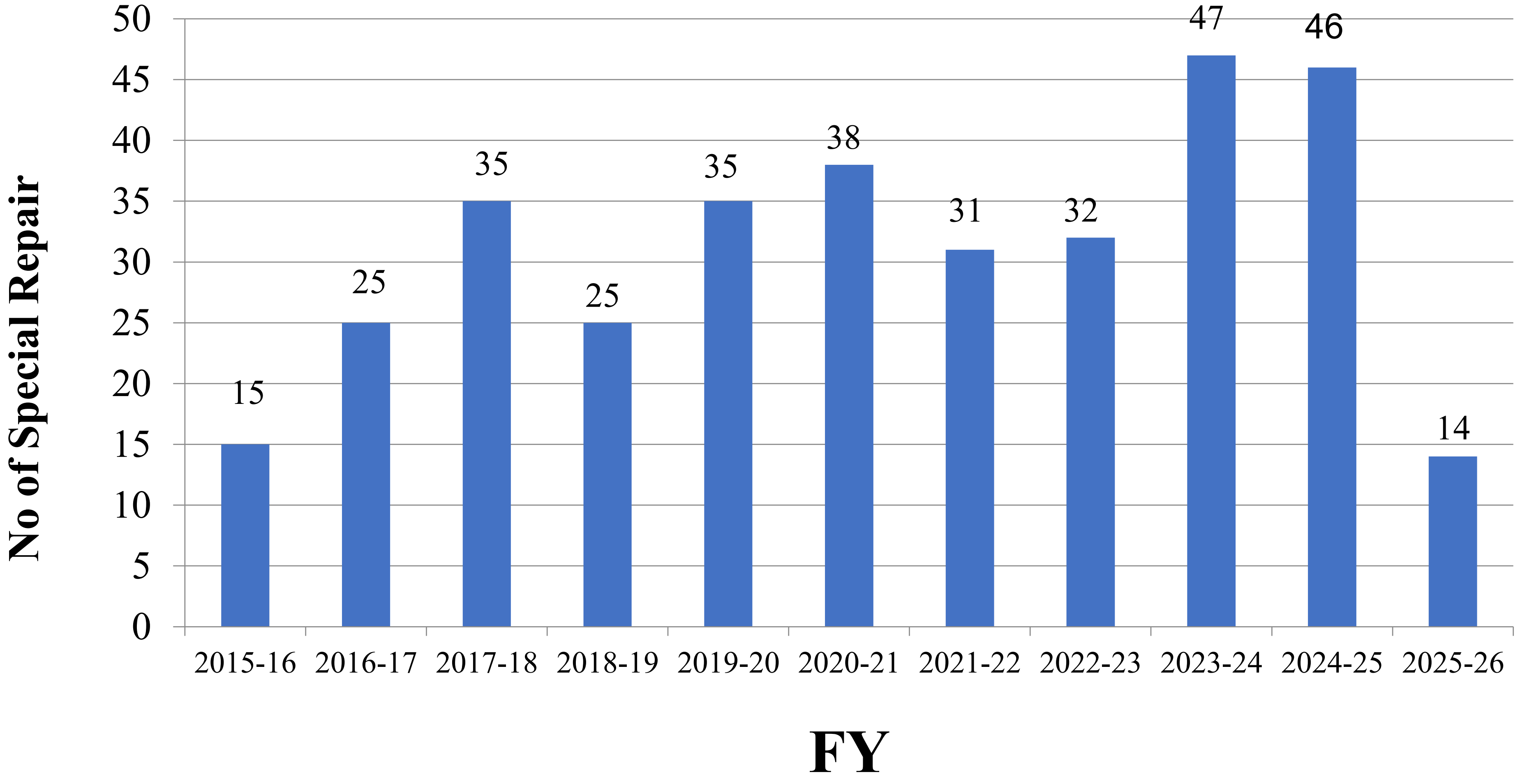
Working Procedure of GOH

Locomotive Receiving → Component Disassembly → Component Cleaning → Component Reconditioning → Component Assembly → Loco Start → Loco Load test → Light & Load Test Run → Loco Painting → Loco Handover to Division.

GOH Out-Turn


No of Out-Turn
21
20
18
16
21
19
19
17
26
01
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23
2023-24
2024-25
2025-26
FY

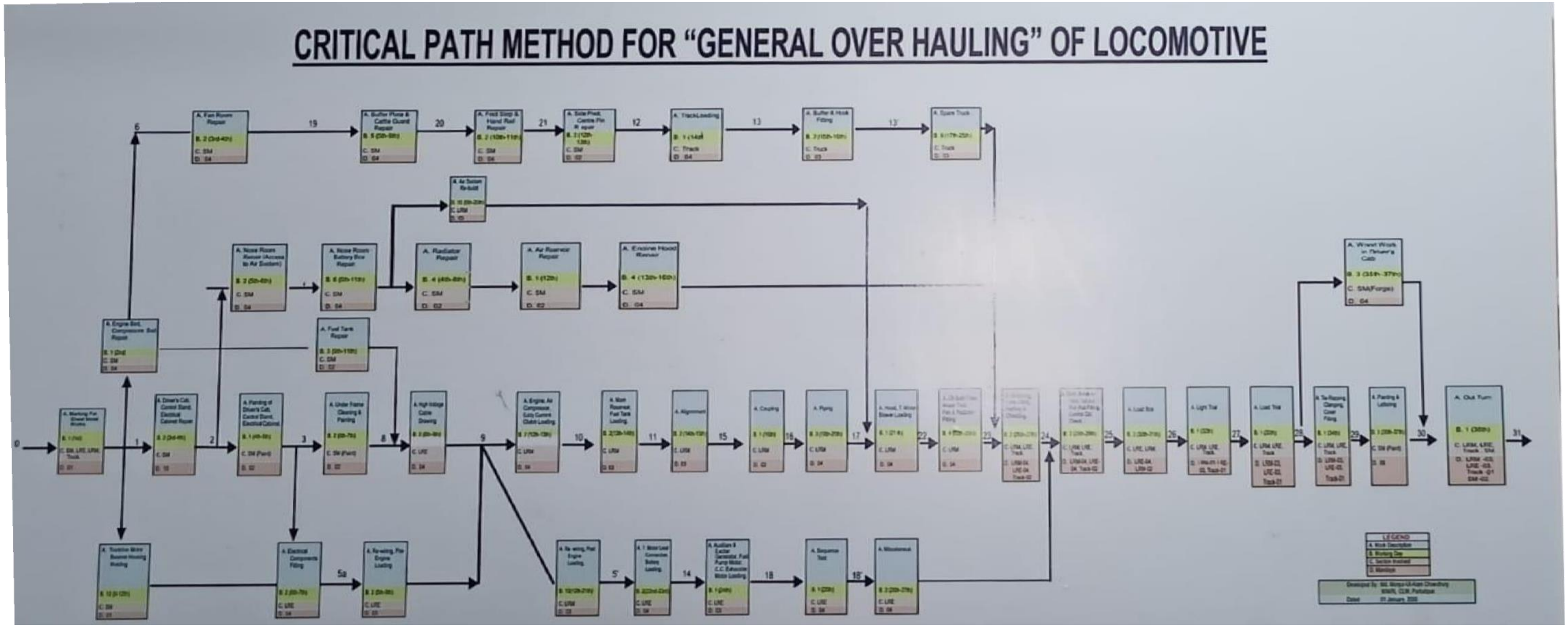
Special Repair



Loco Position of Bangladeh Railway

Gauge	Total no of Locomotive	Running Locomotive	Economic Life Crossed
MG	176	163	113(64.20%)
BG	132	132	41 (31.06%)
Total	308	295	154(50.00%)

Critical Path Method



Sections at CLW

P&P: Planning and progress

LRM: Loco Repair Mechanical

LRE: Loco Repair Electrical

Truck

SM: Sheet Metal

MR: Mechanical Reconditioning

ER: Engine Repair

EM: Electrical Machine

EC: Electrical and electronic component

M & W: Machine and wheel

A& G: Air and governor

LRM: Loco Repair Mechanical

1. Recieve and make D.D.(defect & deficiency) list
2. Disassemble and primary cleaning of mechanical components
3. Send to respective sections
4. Assembly of mechanical components
5. Alignment, piping, fueling, watering, forced lubrication, engine start and break-in test
6. Load box and SFC calculation
7. Light and load trial
8. Handover to respective division

Major machines and plants

A category

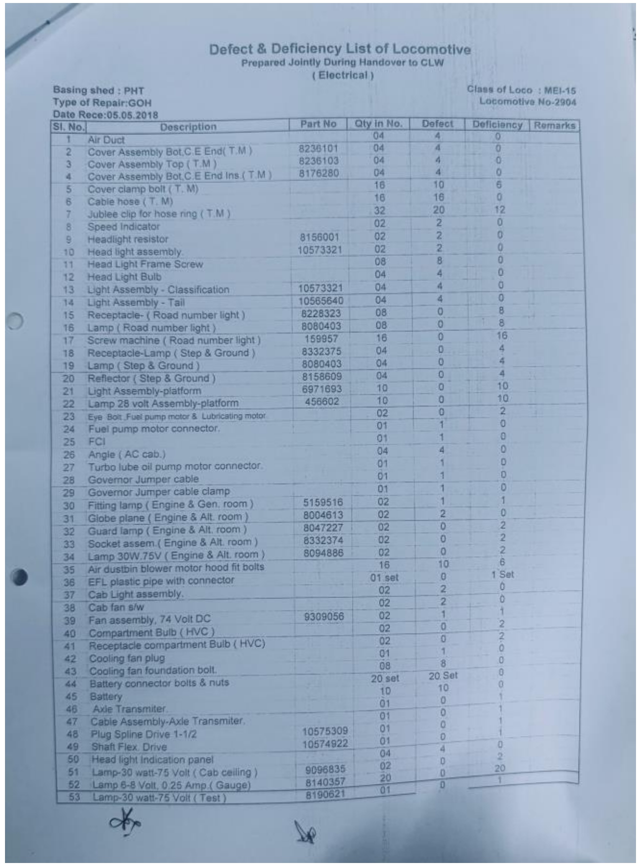
1. Transfer table
2. Overhead crane
3. Lifting stand

B category

1. 3 stage washer for car body filter cleaning
2. Promoted water plant for Water Treatment
3. Lube oil Dispensing Truck for Force Lubrication to Engine
4. Hot water Steam Jet Cleaning machine

LRE: Loco Repair Electrical

1. Receive and making D.D. list of Electrical Components
2. Disassemble and Assemble of electrical component
3. Locomotive Primary Cleaning and Painting
4. Electrical Component Fitting
5. HV and LV Wiring of Loco
6. Different Electrical Circuit Modification
7. Locomotive Fuel Tracker setting
8. Locomotive Vigilance system setting
9. Engine start and check
10. Load test of locomotive
11. Light and Load Trial of locomotive
12. Handover of locomotive



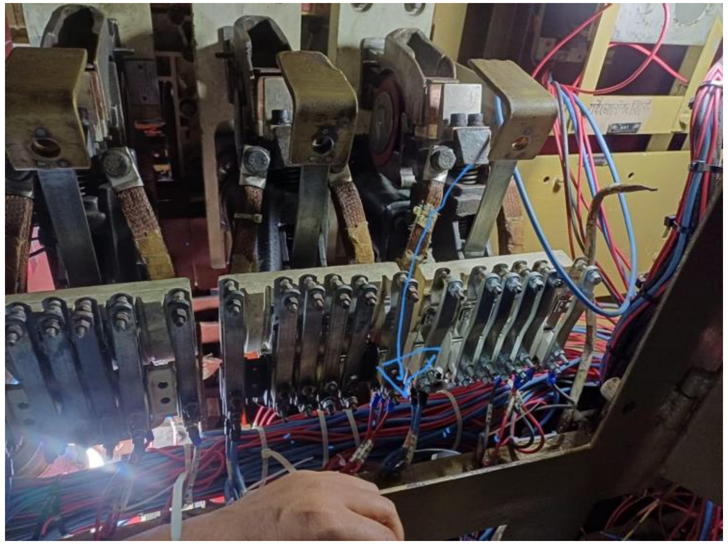
Defect & Deficiency List of Locomotive
Prepared Jointly During Handover to CLW
(Electrical)

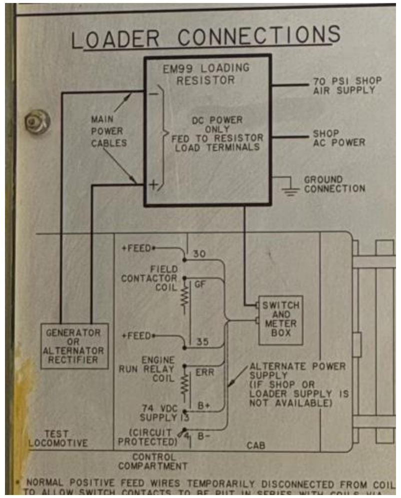
Basing shed : PHT
Type of Repair:GOH
Date Rece:05.05.2018

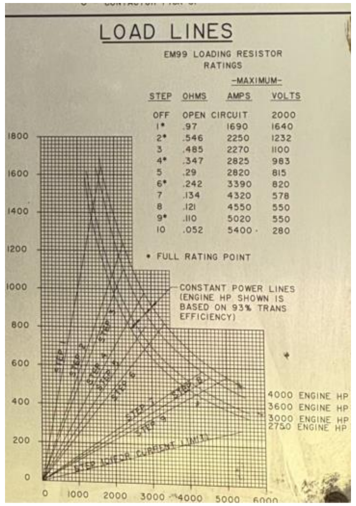
Class of Loco : MEI-15
Locomotive No-2904

Sl. No.	Description	Part No	Qty in No.	Defect	Deficiency	Remarks
1	Air Duct		04	4	0	
2	Cover Assembly Bot.C.E End(T.M)	8236101	04	4	0	
3	Cover Assembly Top (T.M)	8236103	04	4	0	
4	Cover Assembly Bot.C.E End Ins.(T.M)	8176280	04	4	0	
5	Cover clamp bolt (T. M)		16	10	6	
6	Cable hose (T. M)		16	16	0	
7	Jubilee clip for hose ring (T.M)		32	20	12	
8	Speed Indicator		02	2	0	
9	Headlight resistor	8156001	02	2	0	
10	Head light assembly.	10573321	02	2	0	
11	Head Light Frame Screw		08	8	0	
12	Head Light Bulb		04	4	0	
13	Light Assembly - Classification	10573321	04	4	0	
14	Light Assembly - Tail	10565640	04	4	0	
15	Receptacle- (Road number light)	8228323	08	0	8	
16	Lamp (Road number light)	8080403	08	0	8	
17	Screw machine (Road number light)	159957	16	0	16	
18	Receptacle-Lamp (Step & Ground)	8332375	04	0	4	
19	Lamp (Step & Ground)	8080403	04	0	4	
20	Reflector (Step & Ground)	8158609	04	0	4	
21	Light Assembly-platform	6971693	10	0	10	
22	Lamp 28 volt Assembly-platform	456602	10	0	10	
23	Eye Bolt ,Fuel pump motor & Lubricating motor		02	0	2	
24	Fuel pump motor connector.		01	1	0	
25	FCI		01	1	0	
26	Angle (AC cab.)		04	4	0	
27	Turbo lube oil pump motor connector.		01	1	0	
28	Govemor Jumper cable		01	1	0	
29	Govemor Jumper cable clamp		01	1	0	
30	Fitting lamp (Engine & Gen. room)	5159516	02	1	1	
31	Globe plane (Engine & Alt. room)	8004613	02	2	0	
32	Guard lamp (Engine & Alt. room)	8047227	02	0	2	
33	Socket assem.(Engine & Alt. room)	8332374	02	0	2	
34	Lamp 30W.75V (Engine & Alt. room)	8094886	02	0	2	
35	Air dustbin blower motor hood fit bolts		16	10	6	
36	EFL plastic pipe with connector		01 set	0	1 Set	
37	Cab Light assembly.		02	2	0	
38	Cab fan s/w		02	2	0	
39	Fan assembly, 74 Volt DC	9309056	02	1	1	
40	Compartment Bulb (HVC)		02	0	2	
41	Receptacle compartment Bulb (HVC)		02	0	2	
42	Cooling fan plug		01	1	0	
43	Cooling fan foundation bolt.		08	8	0	
44	Battery connector bolts & nuts		20 set	20 Set	0	
45	Battery		10	10	0	
46	Axle Transmiter.		01	0	1	
47	Cable Assembly-Axle Transmiter.		01	0	1	
48	Plug Spline Drive 1-1/2	10575309	01	0	1	
49	Shaft Flex. Drive	10574922	01	0	1	
50	Head light Indication panel		04	4	0	
51	Lamp-30 watt-75 Volt (Cab ceiling)	9096835	02	0	2	
52	Lamp 6-8 Volt, 0.25 Amp.(Gauge)	8140357	20	0	20	
53	Lamp-30 watt-75 Volt (Test)	8190621	01	0	1	

LRE: Loco Repair Electrical







MR: Mechanical Reconditioning

Functions:

- Overhauling of Engine power Assembly
  - Includes Piston, Cylinder head, Connecting Rod, Cylinder liner, Cross Head Lifter(ALCO), Valve Bridge, Governor Drive Gear (ALCO), Overspeed Trip Device(ALCO), Rocker Arm, Yoke etc
- Overhauling of Expressor(Compressor + Exhauster)
  - Includes Complete Dismantling, Maintenance/Replacement of Housing, Crank Shaft, Crank Shaft Bearing, Suction and Discharge Valves, Connecting Rod, Inter Cooler Safety Valve etc
  - Tested in Compressor/Exhauster Test Plant
- Overhauling of Turbocharger
  - Includes Complete Dismantling, Replacement/Maintenance of Parts (Sun Gear, Planetary Gear, Turbine Blades, Diffuser, Soak Back Pump etc)

MR: Mechanical Reconditioning

Functions:

- Overhauling of Water and Lube Oil Pump
- Overhauling of Radiator Cooing fan
- Overhauling of Traction motor blower
- Includes TM Blower(ALCO & GM), Blower Drive Gear (ALCO)
- Repair/Maintenance of Relief/Regulating/Bypass Valve
- Repair/Maintenance of Thermostat Valve, After Cooler Core, Cooling Fan, Eddy Current Clutch

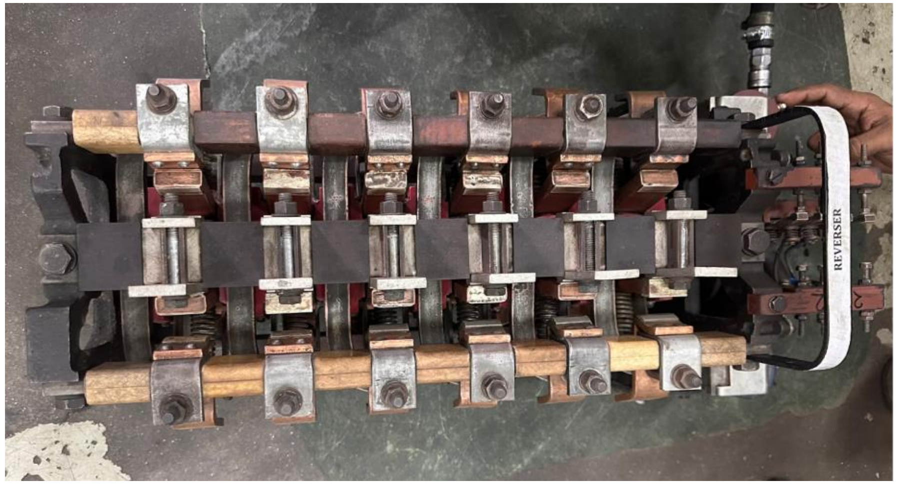
Engine Repair Section

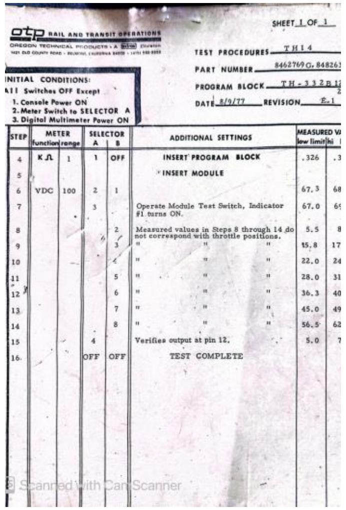
- The primary function of the Engine Repair (ER) Section is the systematic disassembly, cleaning, and preliminary inspection of engine components. The following steps and component-handling procedures are to be strictly adhered to during engine repair activities.
- Engine Cleaning
- Disassembly Process
  - Power Pack (Cylinder Head, Piston with Connecting Rod & Cylinder Liner)
  - Accessories Drive (Accessories Drive Shaft, Lubricating Oil Pump, Water Pump & Associated Piping)
  - Camshaft, Gear Train, All Covers, Auxiliary Generator, AC Core, Turbocharger
  - Crankshaft Assembly (Main Bearing Cap, Crankshaft)
- Testing/Measurement:
  - Bend Test
  - Main Journal Diameter Measurement
  - Crank Pin Diameter Measurement

EC: Electrical and electronic component

Main Function:
Repair, Maintenance & Testing of

- Master controller
- Reverser
- Power contactor
- Relay
- Circuit breaker
- Switch
- Modules, card and battery





EM: Electrical Machine

Check and repair of the following items

- Main generator
- Main Alternator
- Auxiliary Generator
- Exciter Generator
- Fuel Pump motor
- Lube oil Motor
- Crank case exhauster motor
- Eddy current clutch
- Axle alternator
- Techo generator
- Module and reverser motor
- Traction motor
- Cooling fan motor

Machine and Wheel section

Functions

1. Wheel turning, De-axleing
2. Traction Motor axle bore boring
3. production of support Bearing
4. Various type of Bush, intercooler safety valve

Major Machine and Plants

1. Wheel press machine
2. Universal boring and milling machine
3. Crankshaft grinding machine
4. Wheel lathe machine
5. Milling Machine
6. slotting Machine
7. surface grinding machine

Sheet Metal section

Main Function:
Fabrication and repairing large structural components of locomotives including --

1. Traction motor build up
2. Radiator
3. Fuel Tank
4. Main air reservoir
5. Water tank
6. Engine hood
7. Chassis
8. Painting



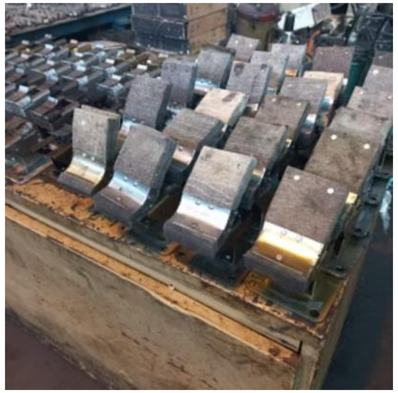
Truck and bogie section

This section mainly focuses on maintenance, repair, and overhauling of locomotive bogies and trucks (the underframe assemblies that carry wheels, axles, and suspension)

It includes the following components-

- Support bearing
- Axle bearing
- Bearing suspension tube
- Wick felt (GM locomotive)
- Truck frame

After dismantling, the axle and wheel are sent to machine section, here Perfect machining is done.



Challenges of CLW

Infrastructure & Machinery:

- Old and Conventional Machinery
- Insufficient precision machining tools

Human Resources:

- Shortage of skilled manpower trained in modern locomotive and workshop technologies.
- High reliance on manual operations instead of automation.

Challenges of CLW

Supply Chain

- Import dependency for critical parts (bearings, injectors, modules).
- No local vendor development for specialized spares.

Process Inefficiencies:

- Lack of Approved Standard Operating Procedures in Each Shop
- Lack of Digital Maintenance Tracking

Strategy-01: Process Optimization & Digitalization

Actions:

- Prepare and Draft the Standard Operating Procedure for each section
- Implement a Computerized Maintenance Management System (CMMS) [barcoding/RFID for tracking maintenance of parts]
- Apply lean manufacturing and 5S principles for workflow efficiency

Expected Outcome:

- Increased Turn Out
- Greater Efficiency in workflow

Strategy-02: Infrastructure Modernization

Actions:

- Installation of modern machinery
- Upgrade Testing Facilities
- Expand workshop and Maximum Utilization of space to separate disassembly and assembly zones.

Expected Outcome:

- Increased Turn Out
- Less Fault During Load Test
- Require Less time for GOH

Strategy-03: Human Resources Development

Actions:

- Organize Advanced Training Program on Modern Locomotives for Section In-charge and Corresponding Staff
- Introduce exchange programs with foreign railways
- Ensure Transfer of Technology

Expected Outcome:

- Highly Efficient Manpower
- Knowledge Sharing
- Capacity Build-up

Strategy-04: Supply Chain Strengthening

Actions:

- LTSA/Framework Contracts with OEMs
- AI inclusion in BRASS

Expected Outcome:

- Spare Parts Availability
- Capacity Enhancement
- Reduce import lead time

Strategy-05: Reverse Engineering

Select the spare parts

Perform Non-Destructive Testing for Material Identification

If Material is available

Measuring the Dimension and Prepare the CAD DWG file of the spare parts

Import the DWG file into CAM software and generate the tool path

Converting the tool path to G-code

Loading G-code to CNC Machine

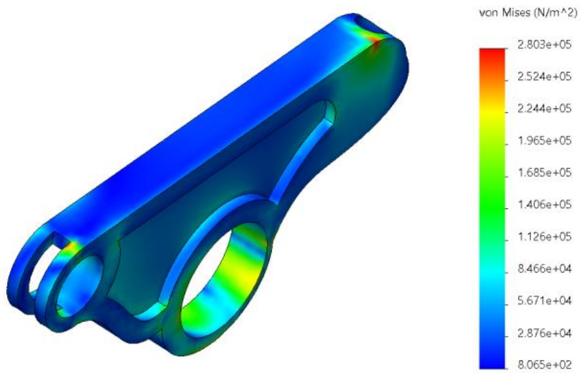
Run the CNC Machine

If Material isn't available

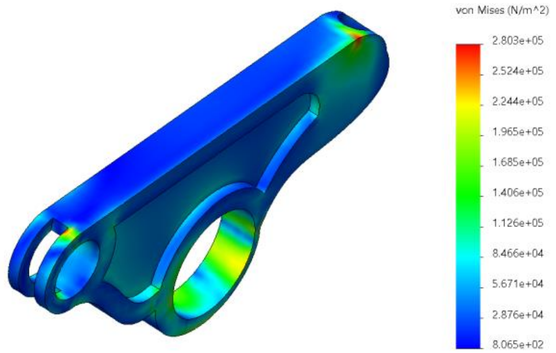
Select a material of the similar composition

Comparative Analysis using Finite Element Method

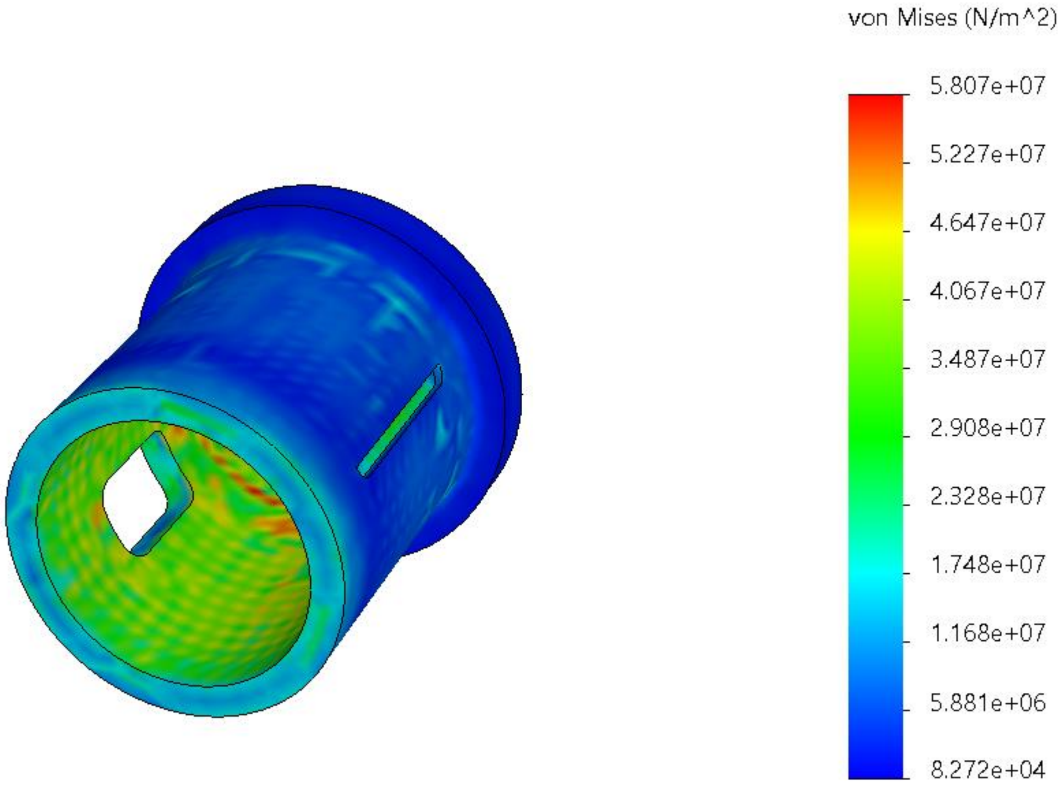
AISI 1045 Steel



AISI 1020 Steel (Cold Drawn and Case Hardened)



Strategy-05: Reverse Engineering



Failure Analysis

Expected Outcome:

- Less Dependency on imported spare parts
- Develop joint ventures with local industries for spare manufacturing
- Capacity Enhancement to build our own locomotive
- Failure Analysis

Strategy-06: Green Workshop

Green Workshop

PV Array Characteristics

PV module		**Inverter**	
Manufacturer	Jinkosolar	Manufacturer	ABB
Model	JKM-610N-78HL4-BDV	Model	PVS-175-TL
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	610 Wp	Unit Nom. Power	175 kWac
Number of PV modules	3864 units	Number of inverters	12 units
Nominal (STC)	2357 kWp	Total power	2100 kWac
Modules	161 Strings x 24 In series	Operating voltage	650-1350 V
At operating cond. (50°C)		Max. power (=>30°C)	185 kWac
Pmpp	2181 kWp	Pnom ratio (DC:AC)	1.12
U mpp	1014 V	Power sharing within this inverter	
I mpp	2151 A		
Total PV power		**Total inverter power**	
Nominal (STC)	2357 kWp	Total power	2100 kWac
Total	3864 modules	Number of inverters	12 units
Module area	10801 m²	Pnom ratio	1.12

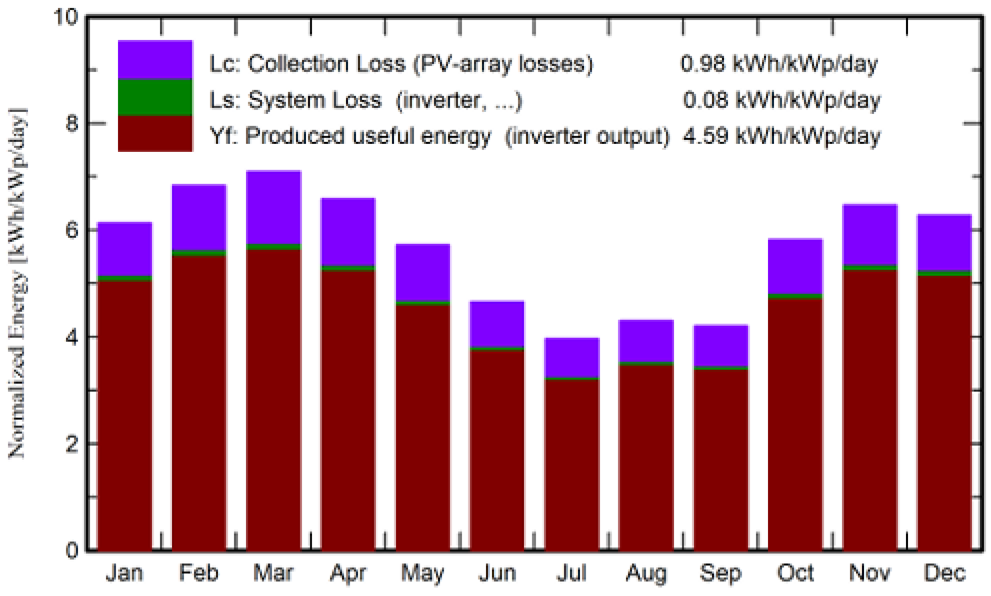
Green Workshop

Main results

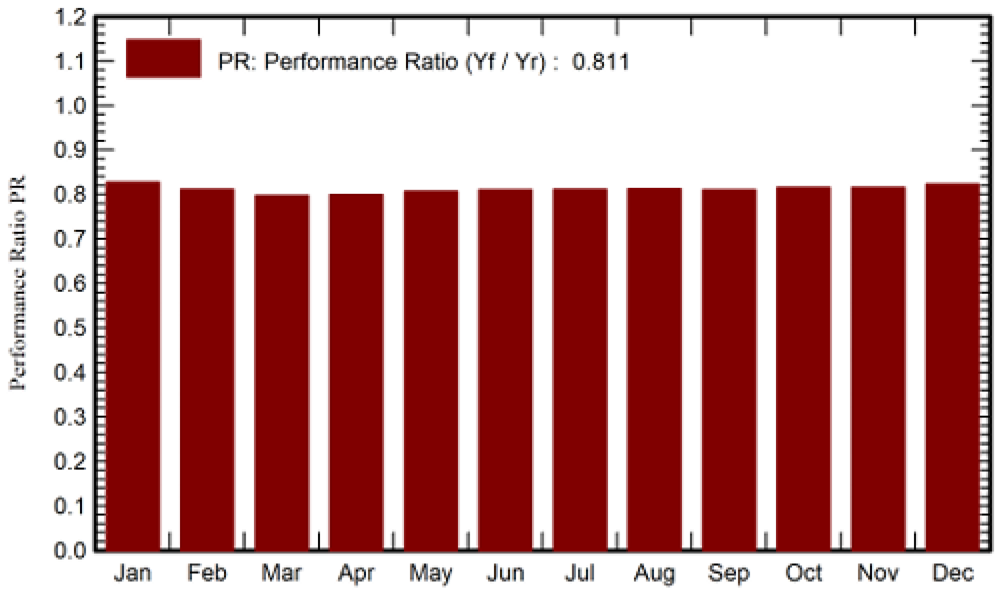
System Production

Produced Energy	3953054 kWh/year	Specific production	1677 kWh/kWp/year
Used Energy	1752000 kWh/year	Performance Ratio PR	81.15 %
		Solar Fraction SF	44.26 %

Normalized productions (per installed kWp)



Performance Ratio PR



Green Workshop

Balances and main results

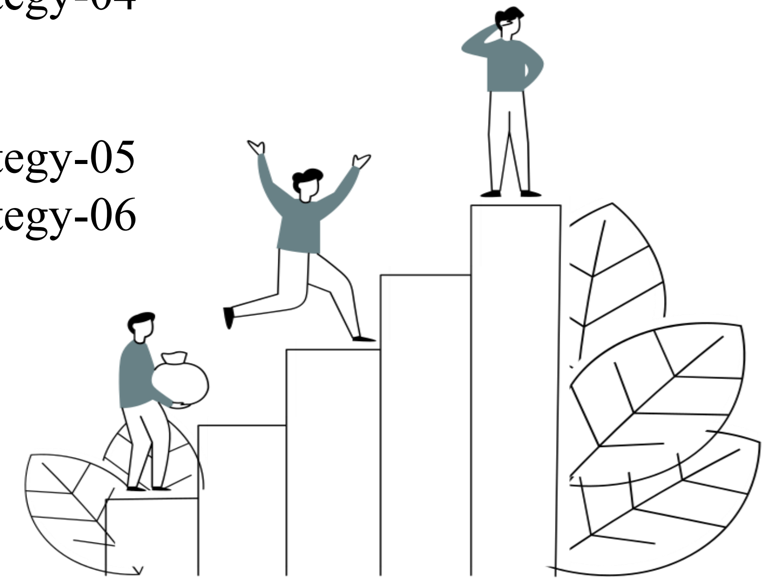
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	133.9	29.14	17.73	190.2	187.5	377747	148800	62182	308780	86618
February	148.4	30.24	20.87	191.3	188.5	372502	134400	58206	307453	76194
March	192.8	42.78	25.08	219.9	216.3	420942	148800	67683	345383	81117
April	194.1	55.80	26.23	197.3	193.3	378721	144000	67178	304778	76822
May	189.7	70.37	26.44	177.3	173.2	343310	148800	70699	266686	78101
June	153.6	75.00	27.29	139.6	136.1	271487	144000	66631	200003	77369
July	133.6	75.64	27.30	122.9	119.6	239222	148800	67986	166942	80814
August	136.7	71.30	27.28	133.4	130.1	259798	148800	67143	188087	81657
September	119.7	61.20	26.25	126.1	123.2	245212	144000	62175	178602	81825
October	148.2	45.57	24.16	180.4	177.6	352942	148800	63235	283274	85565
November	140.4	27.60	21.19	193.9	191.2	379537	144000	60347	312199	83653
December	131.4	24.80	18.81	194.5	191.6	384163	148800	62000	315404	86800
Year	1822.6	609.44	24.07	2066.8	2028.0	4025583	1752000	775464	3177591	976536

Green Workshop

Summary Analysis			
Net Generation	=	3.95	MkWh
Total EPC Cost	=	1241.04	lac taka
Annual Operational Cost	=	97.40	lac taka
Tariff Rate (33 KV Bus)	=	7.045	Tk/Kwh
Financial Analysis Result			
FNPV	=	448.41	lac taka
FBCR	=	1.37	
FIRR	=	18.83%	
Economic Analysis Result			
ENPV	=	1,182.27	lac taka
EBCR	=	2.16	
EIRR	=	33.13%	

Implementation Road Map

Phase	Timeline	Strategy
Short-term (1–2 years)	Drafting SOPs, CMMS introduction, Advanced Training Program	Strategy-01 Strategy-03
Medium-term (3–5 years)	Installation of modern machinery, Upgrading the Testing Facility,, Introducing Exchange Program, LTSA/Framework Contracts with OEMs	Strategy-02 Strategy-03 Strategy-04
Long-term (5–10 years)	Reverse Engineering, Green Workshop	Strategy-05 Strategy-06



Thank you